

# Zichun Hao

California Institute of Technology

 [zichunhao.com](https://zichunhao.com)

 [LinkedIn](#)

 [GitHub](#)

 [Google Scholar](#)

 [ORCID](#)

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## EDUCATION

### California Institute of Technology

Sep 2023 – Present

- Ph.D. candidate in Physics, Division of Physics, Mathematics and Astronomy (PMA)
- Master of Science in Physics, received June 2026
- Thesis advisor: [Prof. Maria Spiropulu](#); research in machine learning and experimental high-energy physics

### University of California, San Diego

Sep 2020 – June 2023

- Bachelor of Science, double major in Physics and Computer Science; Phi Beta Kappa

### University of California, Santa Barbara

Sep 2018 – Aug 2020

- Undergraduate study in Physics

## RESEARCH EXPERIENCE

### Caltech High Energy Physics Group | Graduate Research Assistant

Sep 2023 – Present

- Advisor: [Prof. Maria Spiropulu](#)
- Research at the Compact Muon Solenoid (CMS) Experiment at CERN and on deep learning methods for physics: self-supervised foundation models, neural-network tagger calibration, trigger development, and detector R&D

### Fermi National Accelerator Laboratory (Fermilab) | Visitor

July – Nov 2024; July – Sep 2025

- Supervisor: Prof. Si Xie
- Self-supervised learning methods for high-energy physics; search for double Higgs (HH) production with CMS data at the Large Hadron Collider

### Duarte Lab | Undergraduate Researcher

Jan 2021 – June 2023

- Mentors: [Prof. Javier Duarte](#) and [Dr. Raghav Kansal](#)
- Geometric deep learning for experimental high-energy physics: Lorentz group equivariant jet autoencoders (LGAE, first author) and a ParticleNet-based boosted  $H \rightarrow WW^*$  tagger for the CMS experiment

## RESEARCH PROJECTS

### RINO: Renormalization Group Distillation with No Labels

Nov 2024 – Present

First author

- Matched fully-supervised top tagging with  $100\times$  fewer labels (accuracy  $0.797 \pm 0.015$  finetuned on 1% of labels vs.  $0.770 \pm 0.026$  supervised on 100%); improved out-of-distribution accuracy by 6.7% and AUC on unseen Higgs decay channels by 4.1% (relative, over the strongest baseline) without retraining
- Pretrained transformer encoders with DINO/DINOv2-style self-distillation (iBOT masked-token prediction, KoLeo regularization), using multi-scale  $k_T$ -clustering views as a physically grounded, renormalization-group-inspired source of self-supervised augmentations — ablations attribute an 11.4–14.0% improvement to the multi-scale views
- Selected as a **spotlight** paper at the ML4PS workshop @ NeurIPS 2025 (offered to  $\sim 1\%$  of accepted papers), [arXiv:2509.07486](#); full version under review at NeurIPS 2026

### Search for Non-Resonant HH Production in the $b\bar{b}b\bar{b}$ Final State

Apr 2024 – Present

Core contributor, CMS Collaboration

- One of the main contributors to the boosted  $HH \rightarrow b\bar{b}b\bar{b}$  search, responsible for new BDT trainings, datacard creation, and the final statistical fits
- Developing sensitivity improvements for the next analysis iteration, including a transformer-based approach and data-driven QCD background estimation
- Performed calibration studies of transformer- and graph-based b-taggers (GloParT, ParticleNet), including data/simulation scale factors via the sfBDT method and  $Z \rightarrow b\bar{b}$  control-region cross-checks
- Measured trigger efficiencies and corrections for  $\tau_3/\tau_2$  and tagger score distributions on  $t\bar{t}$  MC samples; processed and managed data and simulation events for the calibration studies
- Presented calibration results at the CMS B-Tagging and Vertexing (BTV) Working Group [\[link\]](#) and CMS Higgs Physics Group [\[link\]](#) meetings
- Contributed to the public analysis [CMS-HIG-24-010](#) (submitted to *Phys. Rev. D*), which sets the most stringent limit on non-resonant HH production to date

## Development of the Barrel Timing Layer (BTL) at the CMS Experiment

Oct 2023 – Present

- Contributed to the quality assurance and quality control (QA/QC) procedure
- Assembled detector modules and trays; performed thermal cooling studies of the module
- Test beam results published in *Nucl. Instrum. Meth. A* **1081**, 170823 (2026)

## Search for Highly Energetic HH Production in the $b\bar{b}VV$ Final State

Mar – Apr 2024

- Optimized event selection based on signal and matching efficiencies; designed a new physics-inspired strategy for choosing signal events
- Analysis summary CMS-PAS-HIG-23-012 made public

## Development of Muon Detector Shower (MDS) Cross Triggers

Nov 2023 – Apr 2024

- Developed the electron+MDS and muon+MDS cross triggers, accepted into the CMS trigger menu in 2023; processed large datasets with parallel workflows
- Trigger system published in *JINST* **21**, P07040 (2026), arXiv:2606.27165

## Lorentz Group Equivariant Jet Autoencoders (LGAE)

Jan 2021 – June 2023

First author; with Prof. Javier Duarte (UC San Diego), Dr. Raghav Kansal (UC San Diego, Fermilab), and Dr. Nadezda Chernyavskaya (CERN)

- Built an autoencoder fully equivariant to the Lorentz group, with a latent space in Lorentz-group irreducible representations, using equivariant message passing with Clebsch–Gordan tensor products
- Outperformed graph- and CNN-based autoencoder baselines on data compression, reconstruction, and unsupervised anomaly detection, with robust performance when trained on only 1% of data
- Published in *Eur. Phys. J. C* **83**, 485 (2023); ~35 citations

## Development of the Tagger for $H \rightarrow WW^*$

June 2021 – Jan 2022

Supported by the UC San Diego Undergraduate Summer Research Award; with Prof. Javier Duarte (UC San Diego), Dr. Raghav Kansal (UC San Diego, Fermilab), Dr. Cristina Mantilla Suarez (Fermilab), and Prof. Si Xie (Caltech)

- Developed a neural-network tagger for  $H \rightarrow WW^*$  decays against QCD backgrounds on the ParticleNet architecture (arXiv:1902.08570) for the CMS experiment
- Related public result: CMS-JME-25-001, particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

## TEACHING

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### Caltech Ph 22 (Computational Physics Laboratory III) | Teaching Assistant

Mar – Jun 2026

- Led weekly lab sessions and taught core AI concepts, from neural-network training to policy-gradient methods; graded coursework and mentored students through their final projects
- One of two TAs who designed the course's final project on reinforcement learning for quantum state preparation: built the Gymnasium environment over a Tavis–Cummings cavity-QED simulator, the REINFORCE reference implementation, and PPO/SAC/TQC baselines (Stable-Baselines3) with MLP/LSTM/Transformer policies
- Designed the project questions, the graded ladder of target quantum states, reward shaping, and multi-seed evaluation requirements

## PUBLICATIONS

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- [Zichun Hao](#), Raghav Kansal, Abhijith Gandrakota, Chang Sun, Jennifer Ngadiuba, Javier Duarte, Maria Spiropulu, “RINO: Renormalization Group Invariance with No Labels,” Nov. 2025, [arXiv:2509.07486](#) ([ML4PS workshop spotlight paper](#), offered to ~1% of accepted papers; [poster](#)). Full version, retitled “RINO: Renormalization Group Distillation with No Labels,” under review at NeurIPS 2026
- Michael Chen, Raghav Kansal, Abhijith Gandrakota, [Zichun Hao](#), Jennifer Ngadiuba, Maria Spiropulu, “An Evaluation of Representation Learning Methods in Particle Physics Foundation Models,” Nov. 2025, [arXiv:2511.12829](#) ([ML4PS paper](#); [poster](#)) — served as a mentor on this work
- Raghav Kansal, Carlos Pareja, [Zichun Hao](#), Javier Duarte, “JetNet: A Python Package for Accessing Open Datasets and Benchmarking Machine Learning Methods in High Energy Physics,” *J. Open Source Softw.* **8**(90), 5789 (2023), doi:[10.21105/joss.05789](#)
- [Zichun Hao](#), Raghav Kansal, Javier Duarte, Nadezda Chernyavskaya, “Lorentz Group Equivariant Autoencoders,” *Eur. Phys. J. C* **83**, 485 (2023), doi:[10.1140/epjc/s10052-023-11633-5](#)
- F. Addesa et al. (incl. [Zichun Hao](#)), “The CMS Barrel Timing Layer: Test Beam Confirmation of Module Timing Performance,” *Nucl. Instrum. Meth. A* **1081**, 170823 (2026), [arXiv:2504.11209](#) — contributor
- M. Abbott et al. (incl. [Zichun Hao](#)), “High Multiplicity Trigger for Long-Lived Particles in CMS Detector,” *JINST* **21**, P07040 (2026), doi:[10.1088/1748-0221/21/07/P07040](#) — contributor
- CMS Collaboration, “Improved results on Higgs boson pair production in the 4b final state,” 2026, [arXiv:2604.27044](#), submitted to *Phys. Rev. D* — core contributor
- CMS Collaboration, “Search for highly energetic double Higgs boson production in the two bottom quark and two vector boson all-hadronic final state,” [CMS-PAS-HIG-23-012](#), CERN (2024) — contributor
- CMS Collaboration, “Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons,” [CMS-PAS-JME-25-001](#), CERN (2025), submitted to *JHEP* — trained the previous ParticleNet-based iteration during undergraduate years

## CONFERENCES & WORKSHOPS

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- [Spotlight talk](#) at the ML4PS workshop, 39th Conference on Neural Information Processing Systems (NeurIPS 2025), on RINO (Dec. 2025, San Diego)
- [ML4Jets 2025](#) on representation learning of jets with physics-informed self-distillations (Aug. 18, 2025, Caltech)
- [APS Global Physics Summit 2025](#) on neural-network tagger calibrations in the search for nonresonant Higgs boson pair production in the four bottom quark decay channel with Run-3 CMS data (Mar. 20, 2025)
- [ML4Jets 2024](#) on Lorentz group equivariant jet autoencoders (Nov. 7, 2024, Paris)
- [APS Far West Section Annual Meeting 2023](#) on Lorentz group equivariant jet autoencoders (Oct. 7, 2023)
- [APS April Meeting 2022](#) on the  $H \rightarrow WW^*$  tagger (Apr. 9–12, 2022)
- [ML4Jets 2021](#) on Lorentz group equivariant jet autoencoders (July 6, 2021)
- [UC San Diego Online Undergraduate Research Symposium 2021](#) on Lorentz group equivariant jet autoencoders (May 26, 2021)

## HONORS & AWARDS

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- [Spotlight paper](#) (first author), ML4PS workshop @ NeurIPS 2025 — offered to ~1% of the workshop’s accepted papers
- [Fermilab LPC](#) Guest and Visitor Award (2024 & 2025)
- [Phi Beta Kappa Honor Society](#) (Apr. 2022)
- UC San Diego Physical Sciences [2022 Dean’s Undergraduate Award for Excellence](#)
- UC San Diego [2021 Undergraduate Summer Research Award](#)
- UC San Diego Provost Honors (Fall 2020, Winter 2021, Spring 2021, Fall 2021, Winter 2022, Spring 2022)
- UC Santa Barbara Dean’s Honors (Winter 2019, Spring 2019, Fall 2019, Winter 2020, Spring 2020)